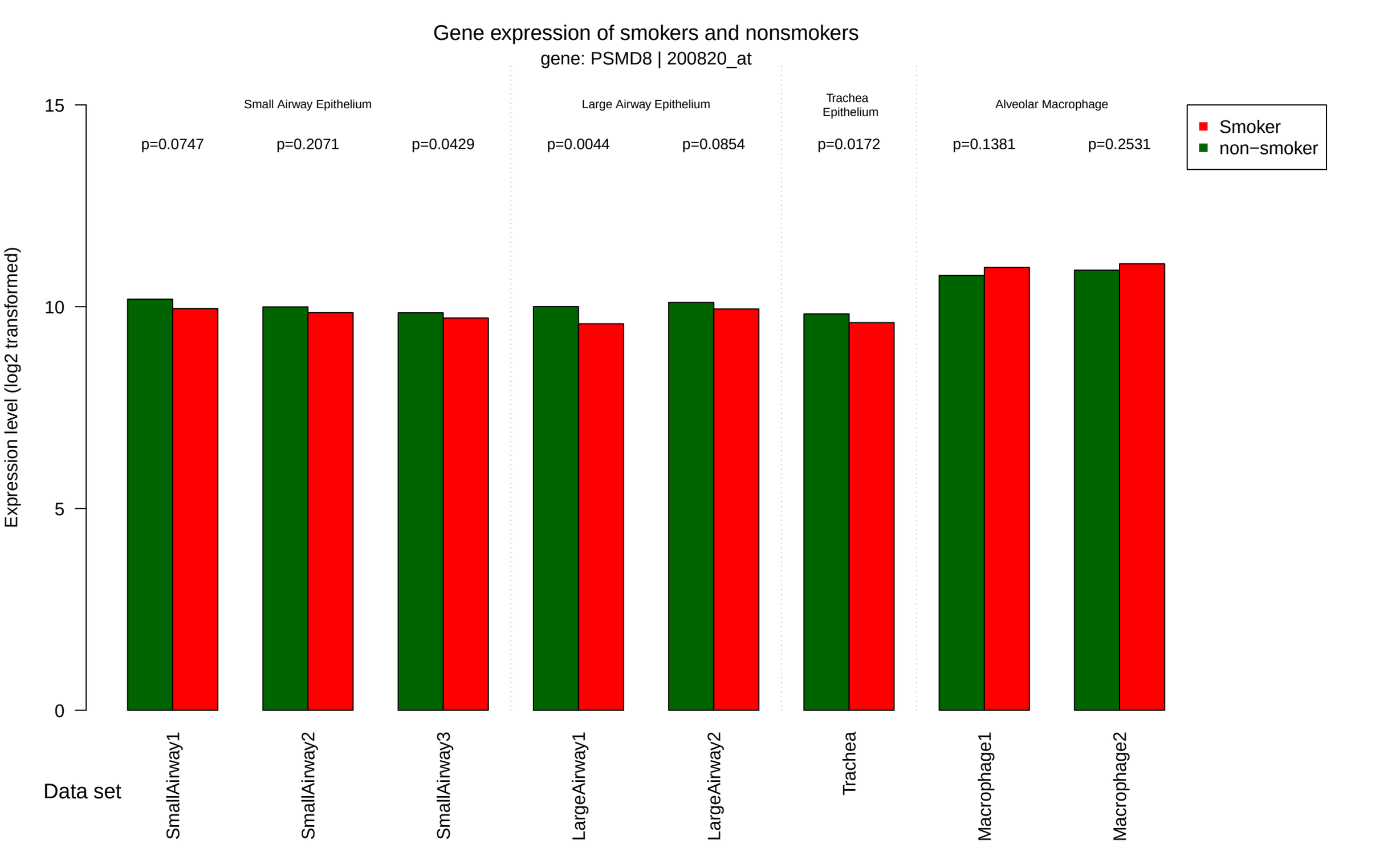
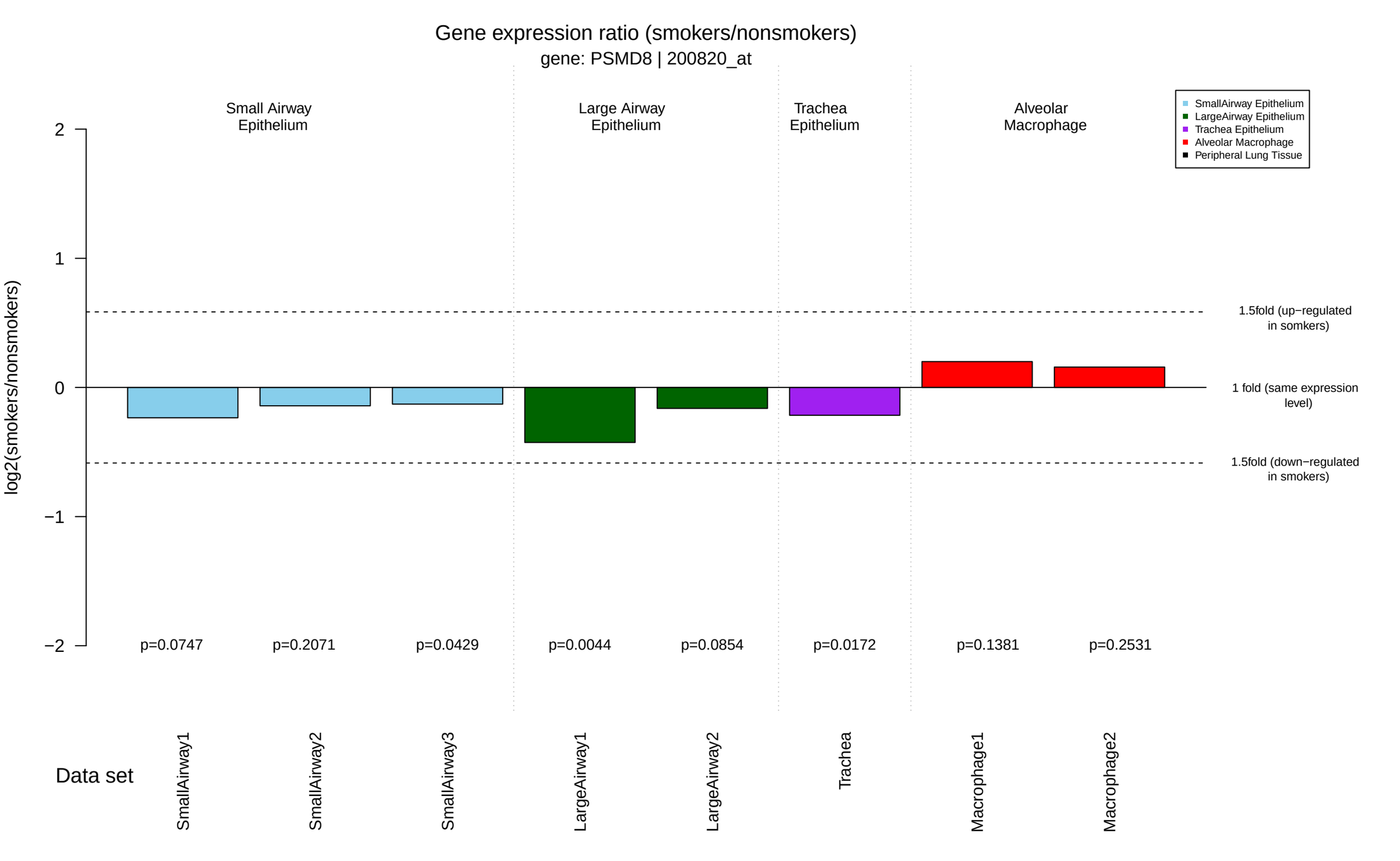
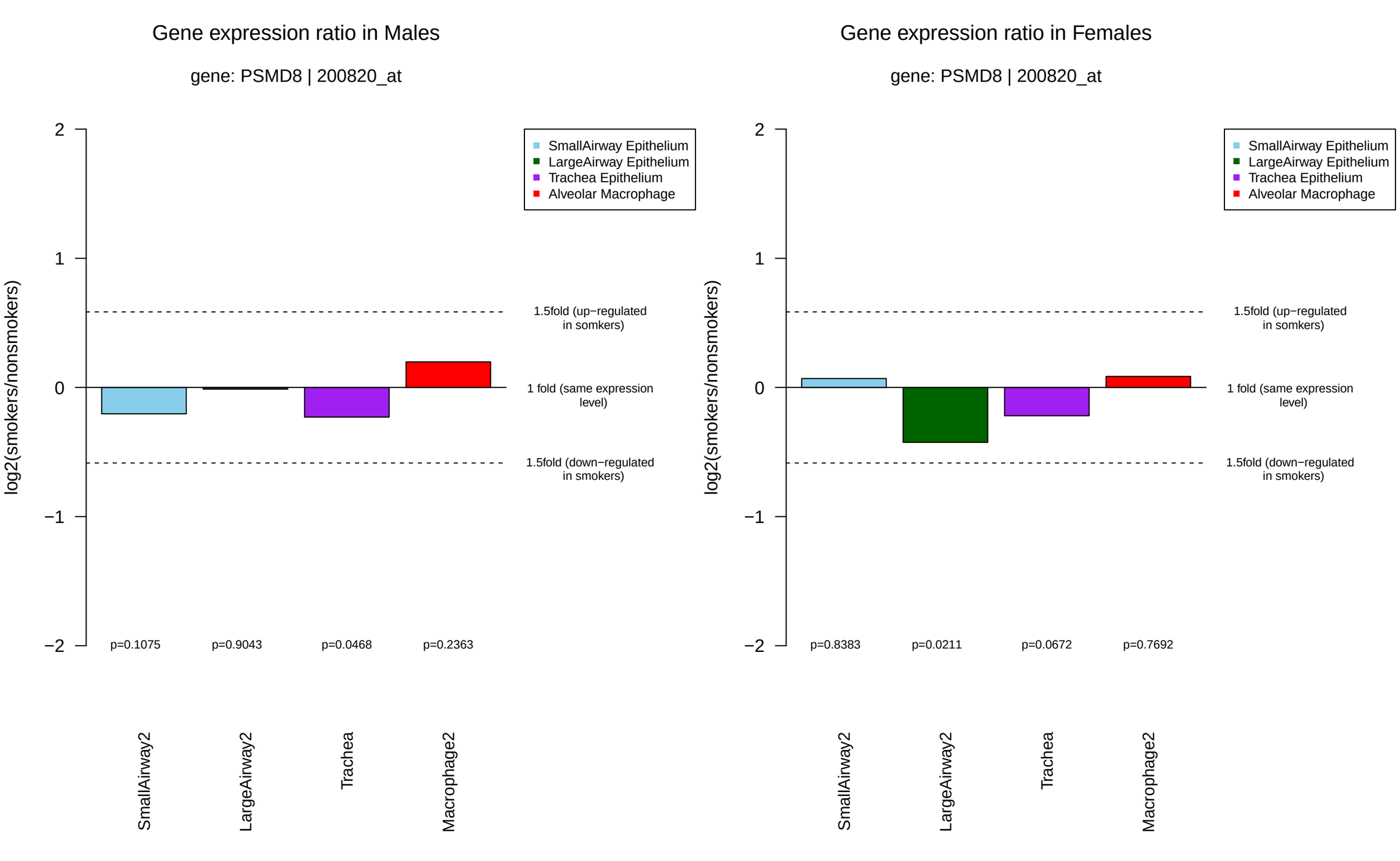
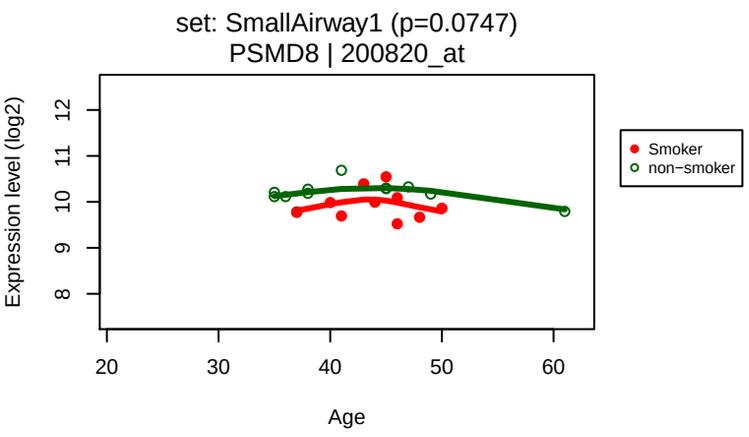
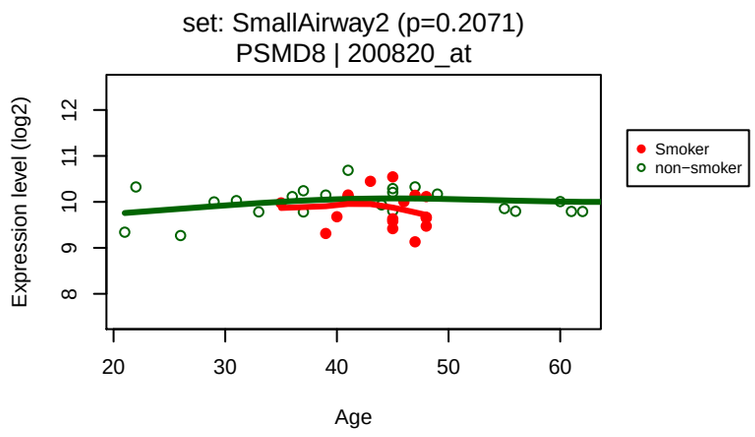
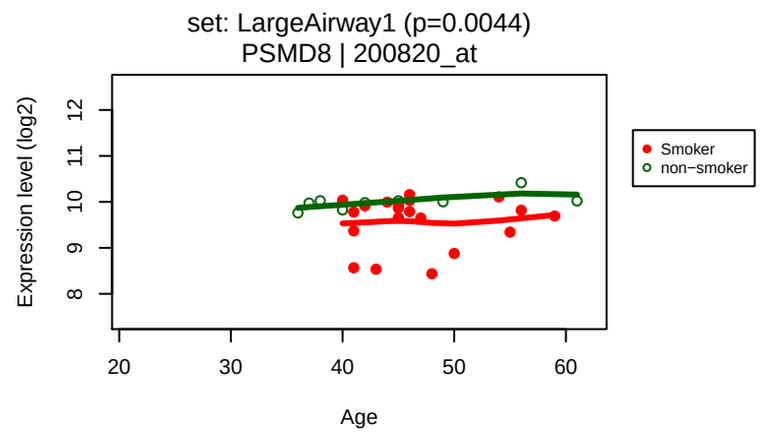

Gene expression of smokers and nonsmokers
gene: PSMD8 | 200820_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0747
p=0.2071
p=0.0429
p=0.0044
p=0.0854
p=0.0172
p=0.1381
p=0.2531
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

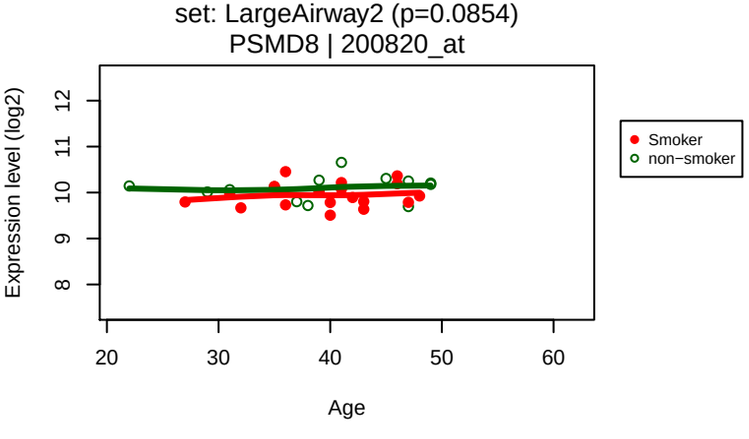

Gene expression ratio (smokers/nonsmokers)
gene: PSMD8 | 200820_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0747
p=0.2071
p=0.0429
p=0.0044
p=0.0854
p=0.0172
p=0.1381
p=0.2531
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

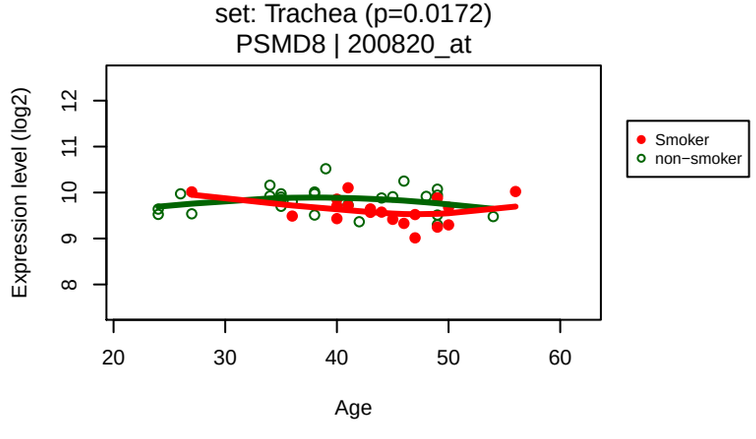

Gene expression ratio in Males
gene: PSMD8 | 200820_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1075
p=0.9043
p=0.0468
p=0.2363
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSMD8 | 200820_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8383
p=0.0211
p=0.0672
p=0.7692
SmallAirway2
LargeAirway2
Trachea
Macrophage2

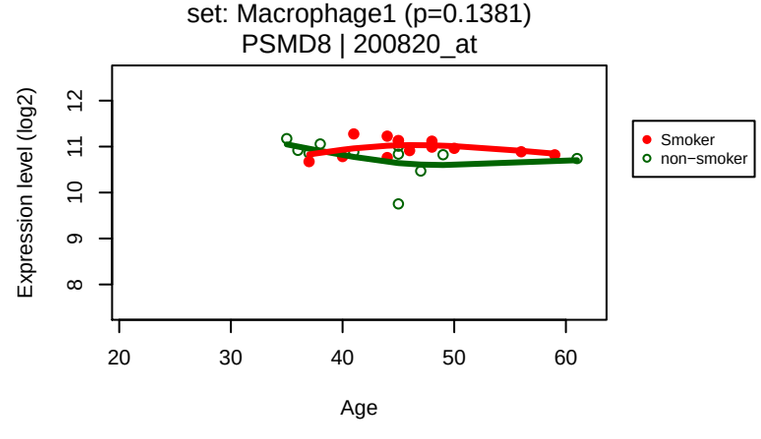

set: SmallAirway1 (p=0.0747)
PSMD8 | 200820_at
Expression level (log2)
Age
Smoker
non−smoker

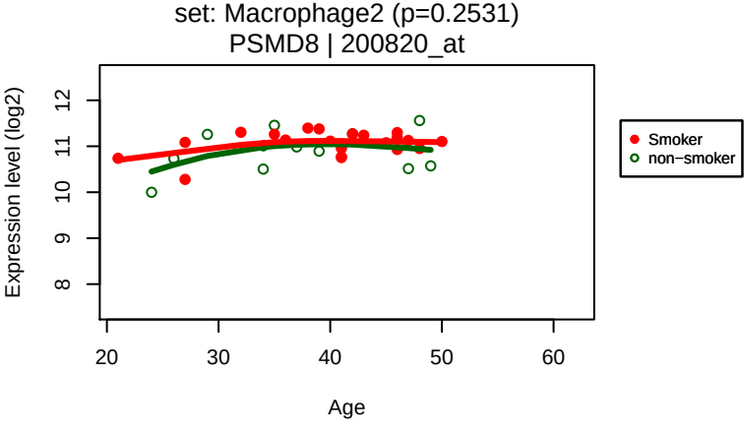

set: SmallAirway2 (p=0.2071)
PSMD8 | 200820_at
Expression level (log2)
Age
Smoker
non−smoker

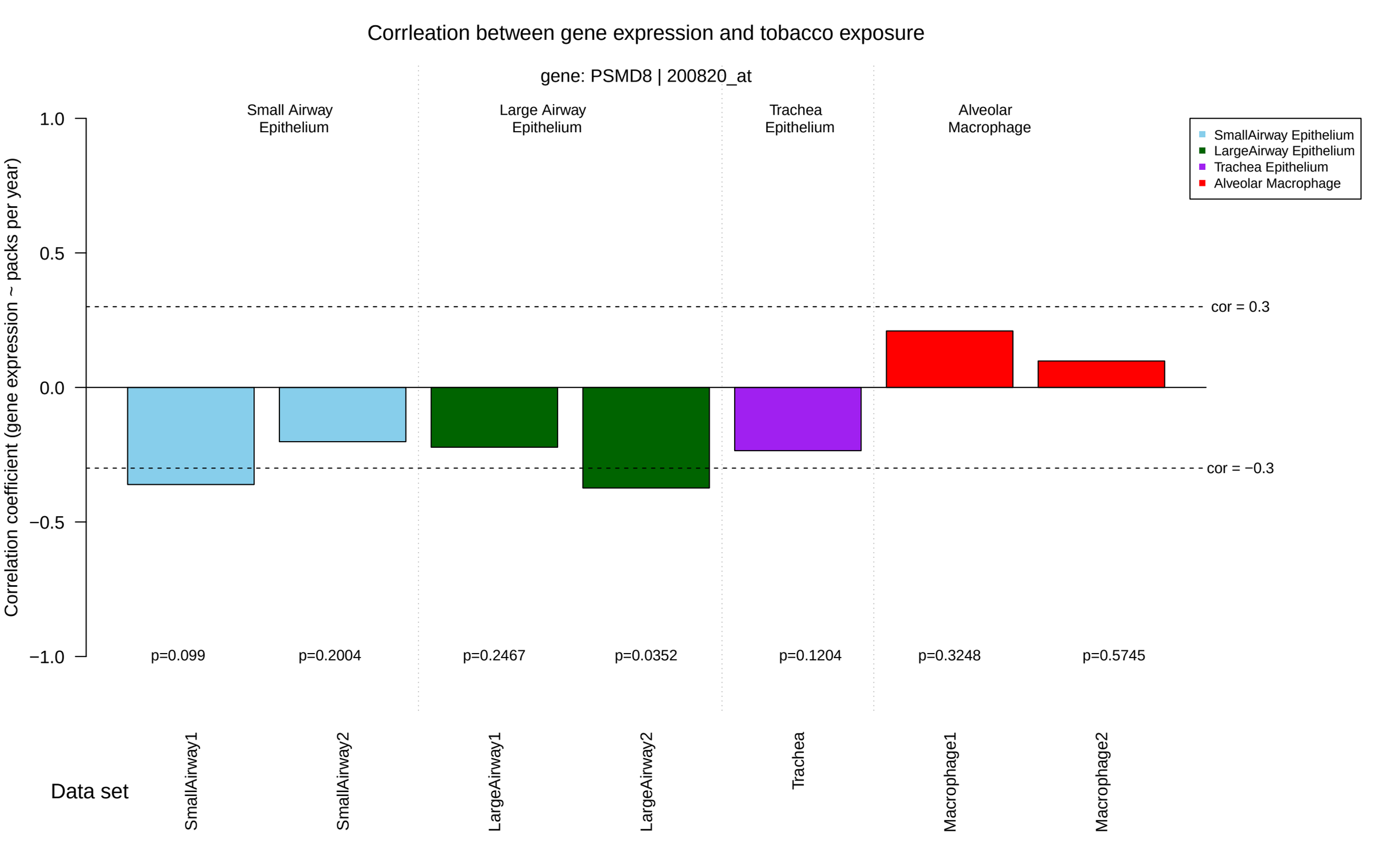

set: LargeAirway1 (p=0.0044)
PSMD8 | 200820_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0854)
PSMD8 | 200820_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0172)
PSMD8 | 200820_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1381)
PSMD8 | 200820_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2531)
PSMD8 | 200820_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSMD8 | 200820_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.099
p=0.2004
p=0.2467
p=0.0352
p=0.1204
p=0.3248
p=0.5745
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2